

On the Minimum Weighted Feedback Vertex Set Problem on the Parallel and Serial Connection of Diamonds

Yen-Lian Chen, Y. L. Wang and Chia-Tung Lee

Information Management

National Chi Nan University, Nantou, Taiwan, ROC

ylchen@alg.csie.ncnu.edu.tw, yuelwang@ncnu.edu.tw, retlee@ncnu.edu.tw

Abstract

In this paper, we introduce two variations of diamond graphs. One is the parallel connection of two diamonds and the other is the serial connection of two diamonds. We shall propose a simple algorithm to solve the minimum weighted feedback vertex problem in the two kinds of diamonds.

A weighted diamond $D_{r,z} = (V, E, w)$ with apices r and z ($r, z \in V$), is an undirected and vertex weighted graph where (i) each $v \in V$ is included in at least one path between r and z and (ii) $D_{r,z} \setminus \{z\}$ is a tree. For example, Figure 2 illustrates three graphs. Figure 2(a) is a weighted diamond graph. Figures 2(b) and 2(c) are not weighted diamond graphs.

1 Introduction

In this paper, we are concerned with the weighted feedback vertex set problem which is defined as follows : We are given an undirected graph $G = (V, E)$ where every vertex $v \in V$ has a weight $c(v)$. A feedback vertex set (FVS) of G is a subset $D \subseteq V$ of vertices such that each cycle in G contains at least one vertex in D , i.e, the subgraph G' induced by the set $V \setminus D$ of vertices is acyclic. The weighted feedback vertex set problem (WFVS) on a graph G is to find an FVS of minimum cost, where the weight of the set is the sum of the weights of its elements.

We shall give an example as follows. Consider Figure 1. There are many feedback vertex sets. For example, $D_1 = \{v_1, v_3\}$ is a feedback vertex set. The cost of D_1 is $c(v_1) + c(v_3) = 32 + 28 = 60$. Another example is $D_2 = \{v_2, v_5\}$ is a feedback vertex set. The cost of D_2 is $c(v_2) + c(v_5) = 24 + 18 = 42$. Finally, let D_3 be $\{v_3\}$. The cost of D_3 is $c(v_3) = 28$. We can see that D_3 is a minimal cost feedback vertex set in this example.

The FVS problem on general graph is NP-hard [5]. For the WFVS problem the best known approximation algorithm has approximation ratio 2 (see, for example, [2,1]). This problem becomes polynomial when addressed on interval graphs [7], convex bipartite graphs [3] and permutation graphs [6]. In this paper, we shall consider a special kind of graphs, namely the parallel and serial connection of two diamonds.

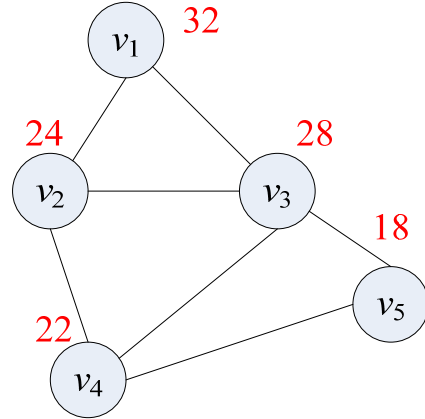


Figure 1. A general graph.

It was proved that the minimum weighted feedback vertex set problem can be solved on diamond graph in linear time by using a dynamic programming algorithm [4]. In this paper, we extend the idea of diamonds by parallel and serial connection of two diamonds. Figure 3 shows a parallel connection of two diamonds and Figure 4 shows a serial connection of two diamonds.

In the following, we shall first give a brief review of the linear algorithms to solve the minimum weighted feedback vertex set problem on diamonds. Then we shall show how we can use this algorithm to find a minimum feedback vertex set on two diamonds connected in either parallel or serial.

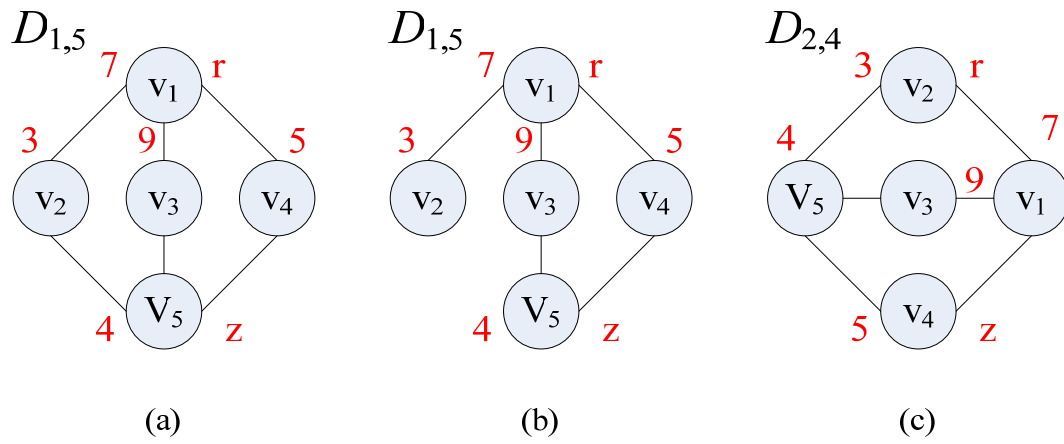


Figure 2. examples.

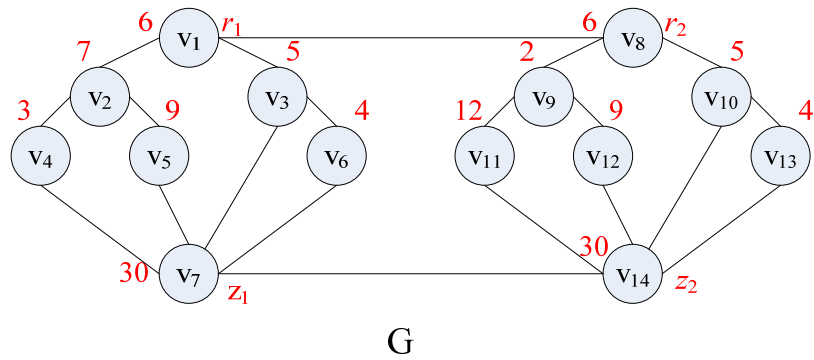


Figure 3. Parallel connection.

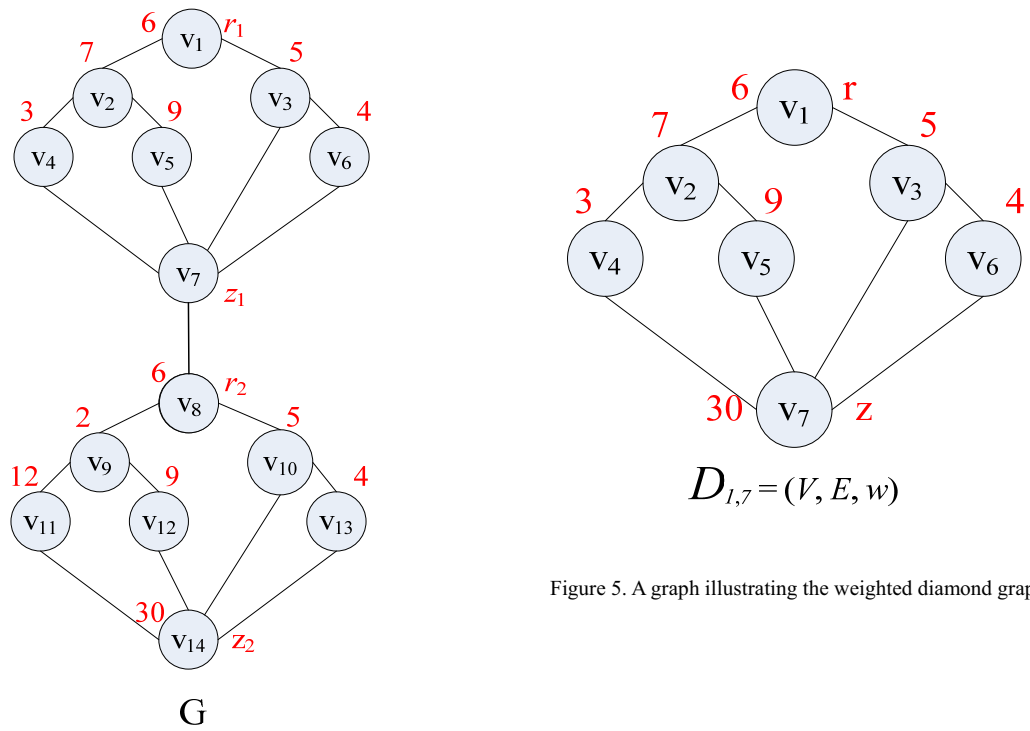


Figure 4. Serial connection.

Figure 5. A graph illustrating the weighted diamond graph.

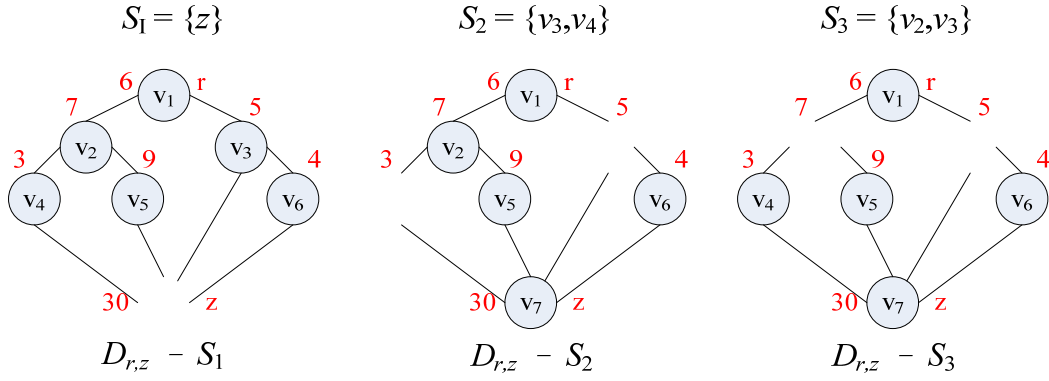


Figure 6. A graph illustrating the solution for weighted diamond graph.

Section 2. Solving the Minimum WFVS Problem on Diamonds

We can see that every weighted feedback vertex set on a diamond $D_{r,z} = (V, E, w)$ is one of the following three kinds of solutions.

- (1) D_1 : A minimum weighted feedback vertex set of $D_{r,z}$ which contains z . In this case, $\{z\}$ itself must be a solution.
- (2) A solution which does not contain z . In this case, there are again two subclasses:
 - (a) D_2 : A minimum weighted feedback vertex set of $D_{r,z}$ which does not contain z and $D_{r,z} - D_2$ contains exactly one path between r and z .
 - (b) D_3 : A minimum weighted feedback vertex set of $D_{r,z}$ which does not contain z and $D_{r,z} - D_3$ contains no path between r and z .

For example, consider Figure 5.

For the diamond graph in Figure 5, there are three solutions for the minimum WFVS problem on this diamond graph and they are shown in Figure 6.

Section 3. The Minimum Weighted Feedback Vertex Set Problem on the Parallel Connection of Two Diamond Graphs

In this section, we consider the parallel connection of two diamonds. Let these two diamonds be denoted as $G_1 = (V_1, E_1, w_1, r_1, z_1)$ and $G_2 = (V_2, E_2, w_2, r_2, z_2)$. Then we define the following :

- (1) D_{11} (respectively, D_{21}) is a minimum feedback vertex set of G_1 (respectively, G_2) under the condition that $D_{r_1, z_1} - D_{11}$ (respectively, $D_{r_2, z_2} - D_{21}$) exactly contains z_1 (respectively, z_2).

- (2) D_{12} (respectively, D_{22}) is a minimum feedback vertex set of G_1 (respectively, G_2) under the condition that $D_{r_1, z_1} - D_{12}$ (respectively, $D_{r_2, z_2} - D_{22}$) exactly contains one path between r_1 and z_1 (respectively, r_2 and z_2).
- (3) D_{13} (respectively, D_{23}) is a minimum feedback vertex set of G_1 (respectively, G_2) under the condition that $D_{r_1, z_1} - D_{13}$ (respectively, $D_{r_2, z_2} - D_{23}$) contains no path between r_1 and z_1 (respectively, r_2 and z_2).

Let parallel diamond be denoted as $G = (V, E, w, r, z)$. Having found $D_{11}, D_{12}, D_{13}, D_{21}, D_{22}$ and D_{23} in G , we can merge them into the following eight solutions :

- (1) $D_{11} \cup D_{21}$ is a minimum feedback vertex set of G . The set exactly contains two lower apexes is z_1 and z_2 .
- (2) $D_{11} \cup D_{22}$ is a minimum feedback vertex set of G . The set exactly contains the lower apex of G_1 and a one path solution of G_2 .
- (3) $D_{11} \cup D_{23}$ is a minimum feedback vertex set of G . The set exactly contains the lower apex of G_1 and a no path solution of G_2 .
- (4) $D_{12} \cup D_{21}$ is a minimum feedback vertex set of G . The set exactly contains a one path solution of G_1 and the lower apex of G_2 .
- (5) $D_{12} \cup D_{23}$ is a minimum feedback vertex set of G . The set exactly contains a one path solution of G_1 and a no path solution of G_2 .
- (6) $D_{13} \cup D_{21}$ is a minimum feedback vertex set of G . The set exactly contains a no path solution of G_1 and lower apex of G_2 .
- (7) $D_{13} \cup D_{22}$ is a minimum feedback vertex set of G . The set exactly contains a no path solution of G_1 and a one path solution of G_2 .



Figure 7. An example to illustrate the Parallel Connection of diamond graph.

- (8) $D_{13} \cup D_{23}$ is a minimum feedback vertex set of G . The set exactly contains a no path solution of G_1 and a no path solution of G_2 .

It is obvious that our minimum weighted feedback vertex set solution of parallel connection of two diamonds must be one of the above solutions. Thus to solve the problem, we may use dynamic programming method introduced in [4] to find D_{11} , D_{12} , D_{13} , D_{21} , D_{22} , and D_{23} , merge them into eight solutions and output the one with the minimum weight as our final solution.

We give an example as Figure 7 and calculate as follows.

At first we can obtain the sets of vertices in D_{11} , D_{12} , D_{13} , D_{21} , D_{22} , and D_{23} and which are $\{7\}$, $\{4, 3\}$, $\{2, 3\}$, $\{14\}$, $\{9, 13\}$, and $\{9, 10\}$ and their costs are 30, 8, 12, 30, 6, and 7, respectively.

Finally, we can merge them into the following eight solutions and we select the minimum weighed set from eight solutions as minimum weighted feedback vertex set. The eight solutions are as follows:

- (1) $D_{11} \cup D_{21} = \{7, 14\}$, cost 60.
- (2) $D_{11} \cup D_{22} = \{7, 9, 13\}$, cost 36.
- (3) $D_{11} \cup D_{23} = \{7, 9, 10\}$, cost 37.
- (4) $D_{12} \cup D_{21} = \{3, 4, 14\}$, cost 38.
- (5) $D_{12} \cup D_{23} = \{3, 4, 9, 10\}$, cost 15.
- (6) $D_{13} \cup D_{21} = \{2, 3, 14\}$, cost 42.
- (7) $D_{13} \cup D_{22} = \{2, 3, 9, 13\}$, cost 18.
- (8) $D_{13} \cup D_{23} = \{2, 3, 9, 10\}$, cost 19.

We will select $\text{Min}\{\{7, 14\}, \{7, 9, 13\}, \{7, 9, 10\}, \{3, 4, 14\}, \{3, 4, 9, 10\}, \{2, 3, 14\}, \{2, 3, 9, 13\}, \text{ and } \{2, 3, 9, 10\}\}$. The minimum feedback vertex set is $\{3, 4, 9, 10\}$ and its cost is 15.

Section 4. The Minimum Weighted Feedback Vertex Set Problem on the Serial Connection of Two Diamonds

In this section, we consider the serial connection of two diamond graphs. Similarly, we can found D_{11} , D_{12} , D_{13} , D_{21} , D_{22} and D_{23} in serial

connection of two diamond graphs and we can merge them into the following nine solutions :

- (1) $D_{11} \cup D_{21}$ is a minimum feedback vertex set of G . The set exactly contains two lower apexes which is z_1 and z_2 .
- (2) $D_{11} \cup D_{22}$ is a minimum feedback vertex set of G . The set exactly contains the lower apex of G_1 and one path solution of G_2 .
- (3) $D_{11} \cup D_{23}$ is a minimum feedback vertex set of G . The set exactly contains the lower apex of G_1 and no path solution of G_2 .
- (4) $D_{12} \cup D_{21}$ is a minimum feedback vertex set of G . The set exactly contains one path solution of G_1 and the lower apex of G_2 .
- (5) $D_{12} \cup D_{23}$ is a minimum feedback vertex set of G . The set exactly contains one path solution of G_1 and no path solution of G_2 .
- (6) $D_{13} \cup D_{21}$ is a minimum feedback vertex set of G . The set exactly contains no path solution of G_1 and lower apex of G_2 .
- (7) $D_{13} \cup D_{22}$ is a minimum feedback vertex set of G . The set exactly contains no path solution of G_1 and one path solution of G_2 .
- (8) $D_{13} \cup D_{23}$ is a minimum feedback vertex set of G . The set exactly contains no path solution of G_1 and no path solution of G_2 .

$D_{12} \cup D_{22}$ is a minimum feedback vertex set of G . The set exactly contains one path solution of G_1 and one path solution of G_2 .

It is obvious that our minimum weighted feedback vertex set solution of serial connection of two diamonds must be one of the above solutions. Thus to solve the problem, we may use dynamic programming method in [4] to find D_{11} , D_{12} , D_{13} , D_{21} , D_{22} , and D_{23} . Then merge them into nine solutions and produce the minimum weighted one as our final solution. We

give an example as Figure 8 and calculate as follows.

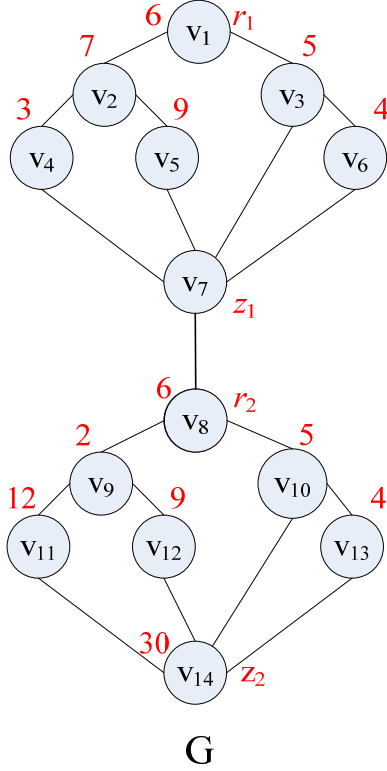


Figure 8 Serial connection.

At first we can obtain the sets of vertices in D_{11} , D_{12} , D_{13} , D_{21} , D_{22} , and D_{23} and which are $\{7\}$, $\{4, 3\}$, $\{2, 3\}$, $\{14\}$, $\{9, 13\}$, and $\{9, 10\}$ and their costs are 30, 8, 12, 30, 6, and 7, respectively.

Finally, we can merge them into the following nine solutions and we select the minimum weighed set from eight solutions the as minimum weighed feedback vertex set.

- (1) $D_{11} \cup D_{21} = \{7, 14\}$, cost 60.
- (2) $D_{11} \cup D_{22} = \{7, 9, 13\}$, cost 36.
- (3) $D_{11} \cup D_{23} = \{7, 9, 10\}$, cost 37.
- (4) $D_{12} \cup D_{21} = \{3, 4, 14\}$, cost 38.
- (5) $D_{12} \cup D_{23} = \{3, 4, 9, 10\}$, cost 15.
- (6) $D_{13} \cup D_{21} = \{2, 3, 14\}$, cost 42.
- (7) $D_{13} \cup D_{22} = \{2, 3, 9, 13\}$, cost 18.
- (8) $D_{13} \cup D_{23} = \{2, 3, 9, 10\}$, cost 19.
- (9) $D_{12} \cup D_{22} = \{4, 3, 9, 13\}$, cost 14.

We will select $\text{Min}\{\{7, 14\}, \{7, 9, 13\}, \{7, 9, 10\}, \{3, 4, 14\}, \{3, 4, 9, 10\}, \{2, 3, 14\}, \{2, 3, 9, 13\}, \{2, 3, 9, 10\}, \text{ and } \{4, 3, 9, 13\}\}$. The minimum feedback vertex set is $\{3, 4, 9, \text{ and } 13\}$, cost 14.

Section 5. Conclusion

In this paper, we have presented the parallel and serial connection of diamond graphs and shown that it is possible to solve WFVP on them in linear time. In the future, we shall try to extend our results to tackle the series and parallel graphs [8, 9, 10, 11] constructed out of diamond graphs.

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